

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo

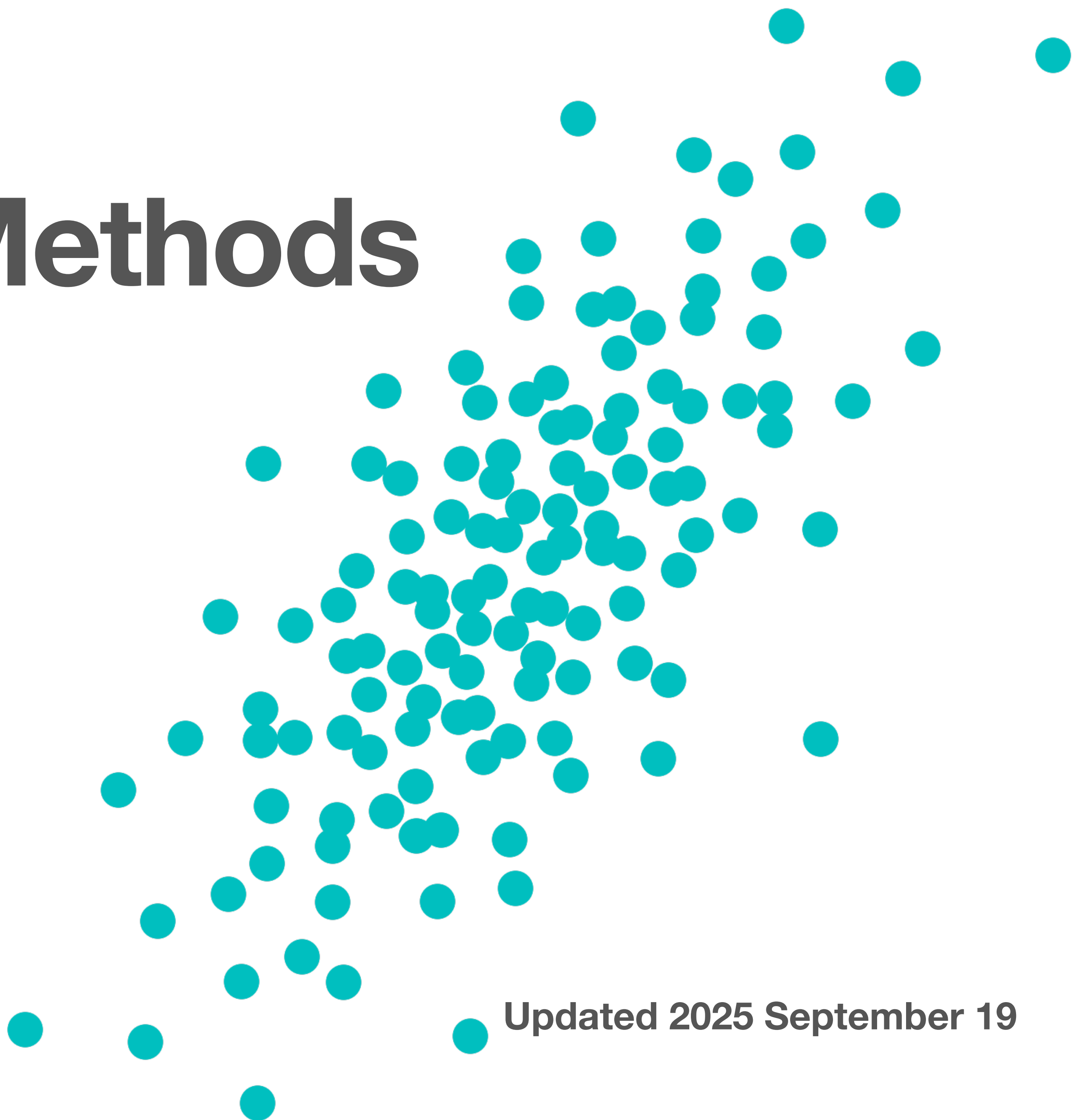
Improving Efficiency

Selected Topics

Git website and repository
Canvas

Fred Hickernell, Fall 2025

Updated 2025 September 19

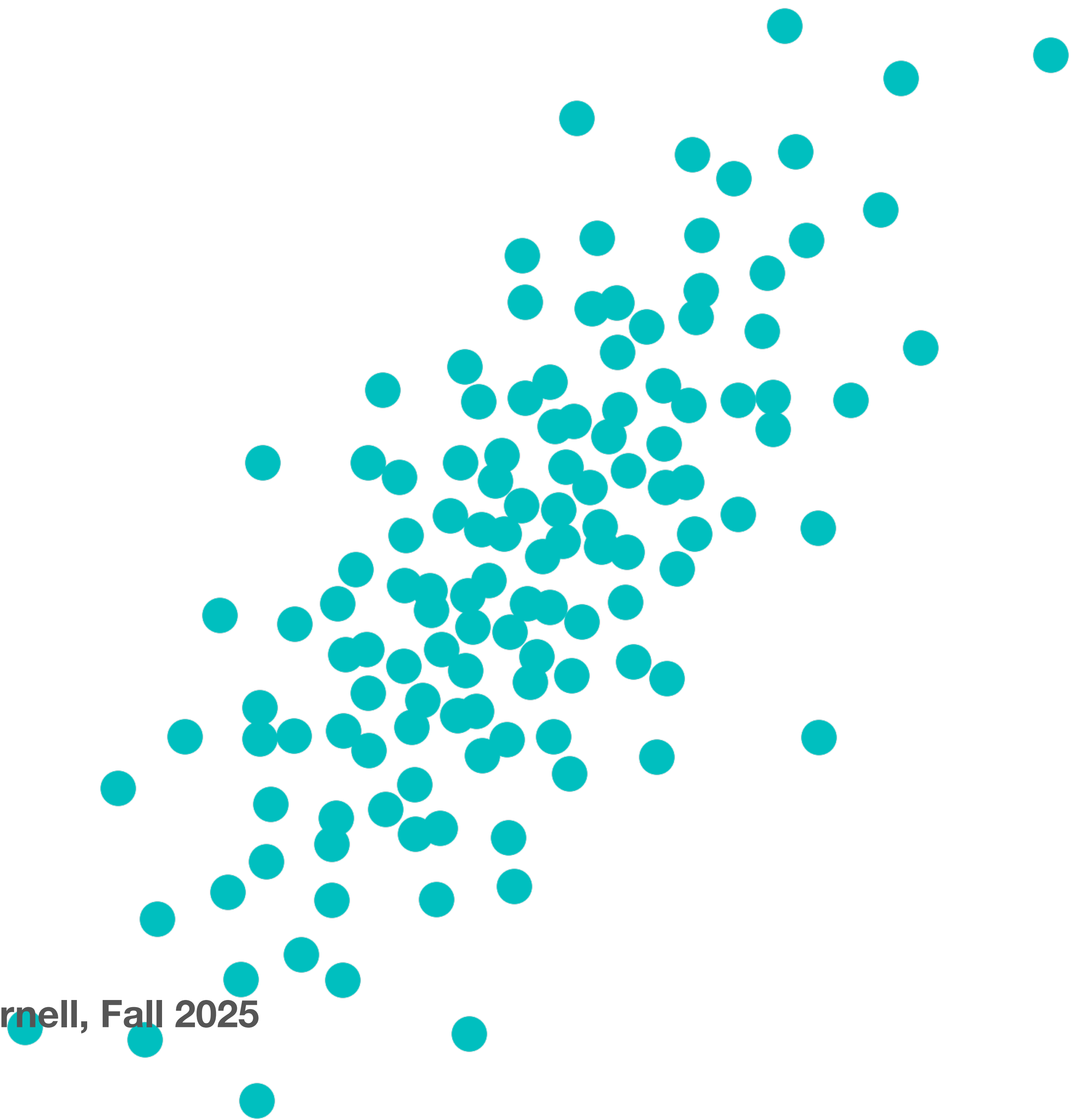


Introduction

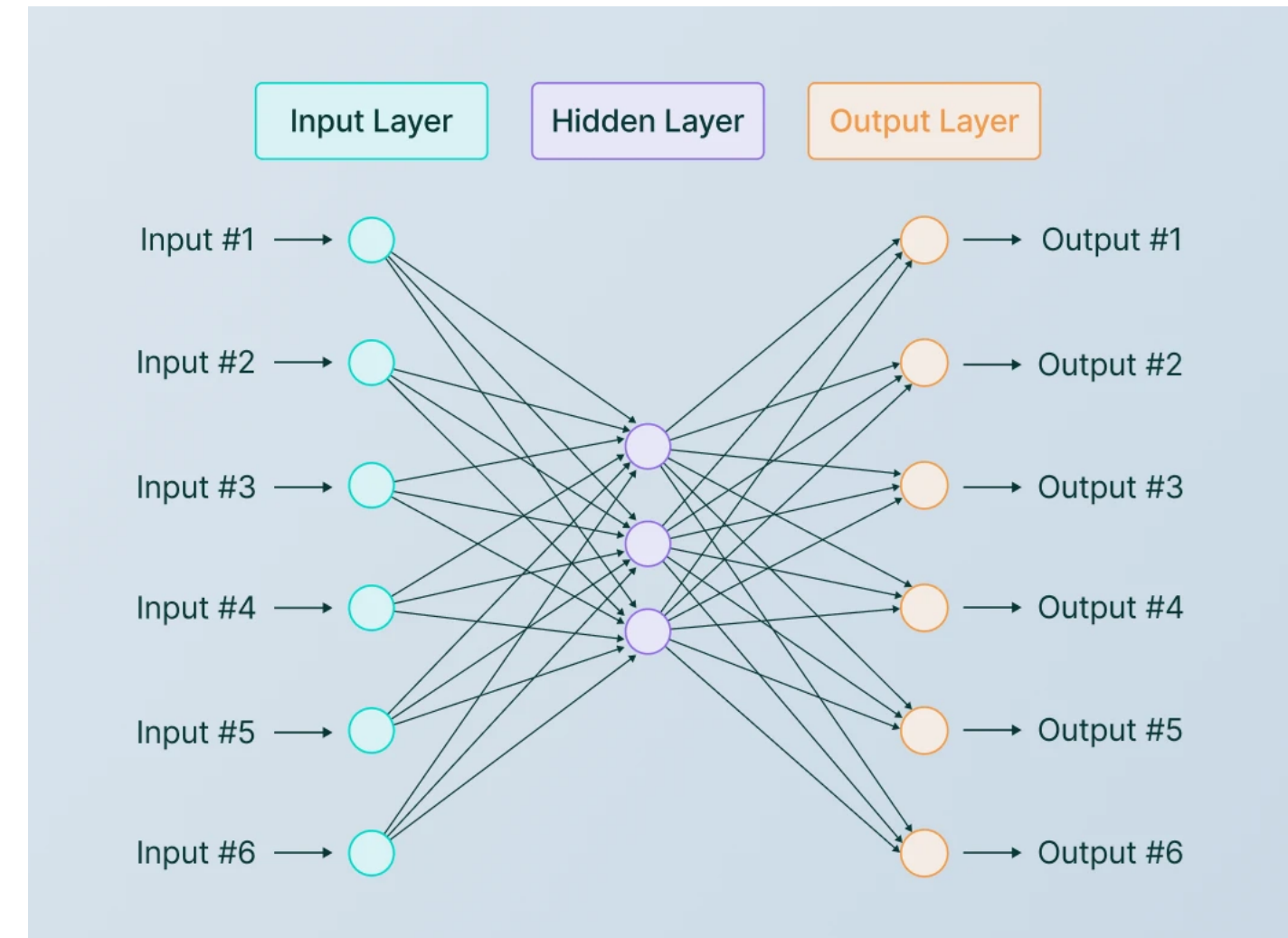
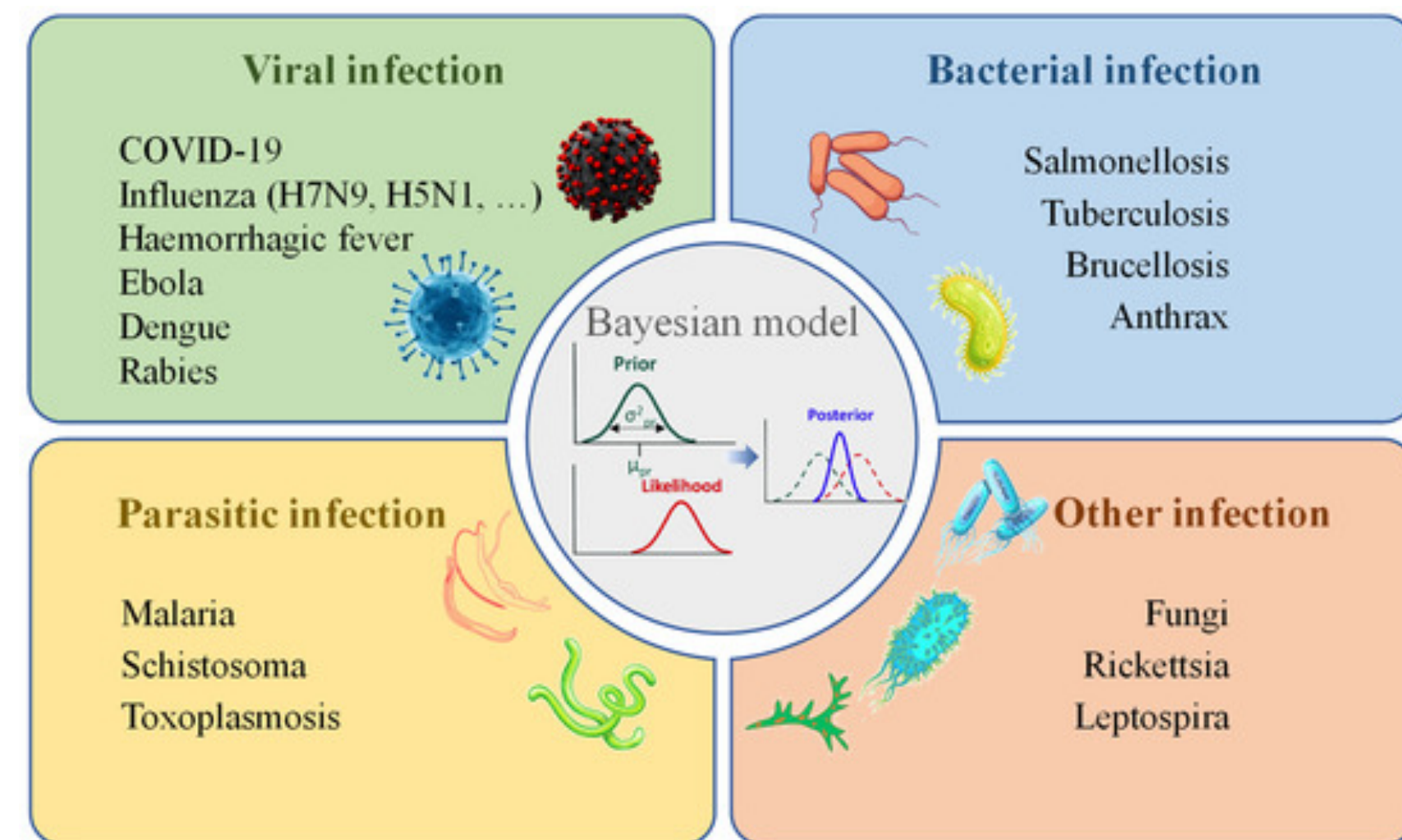
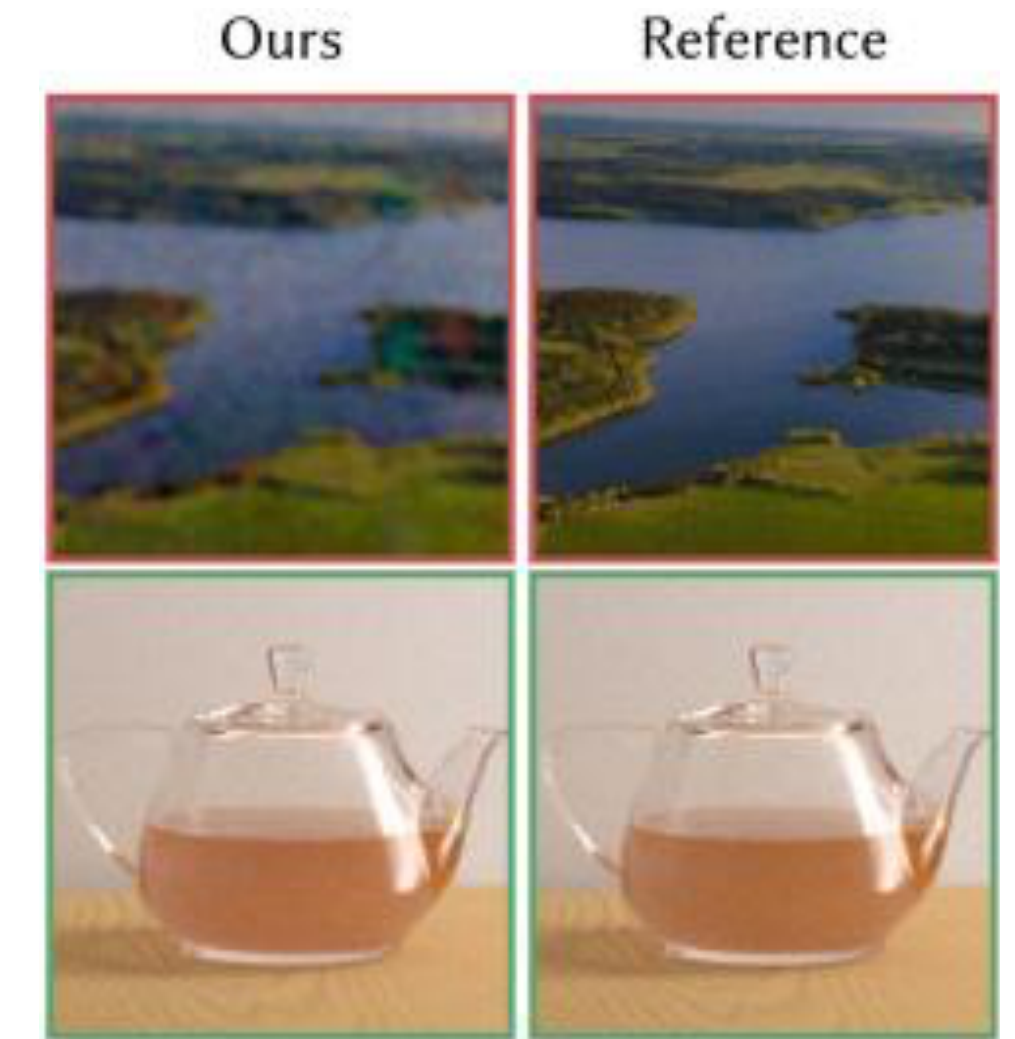
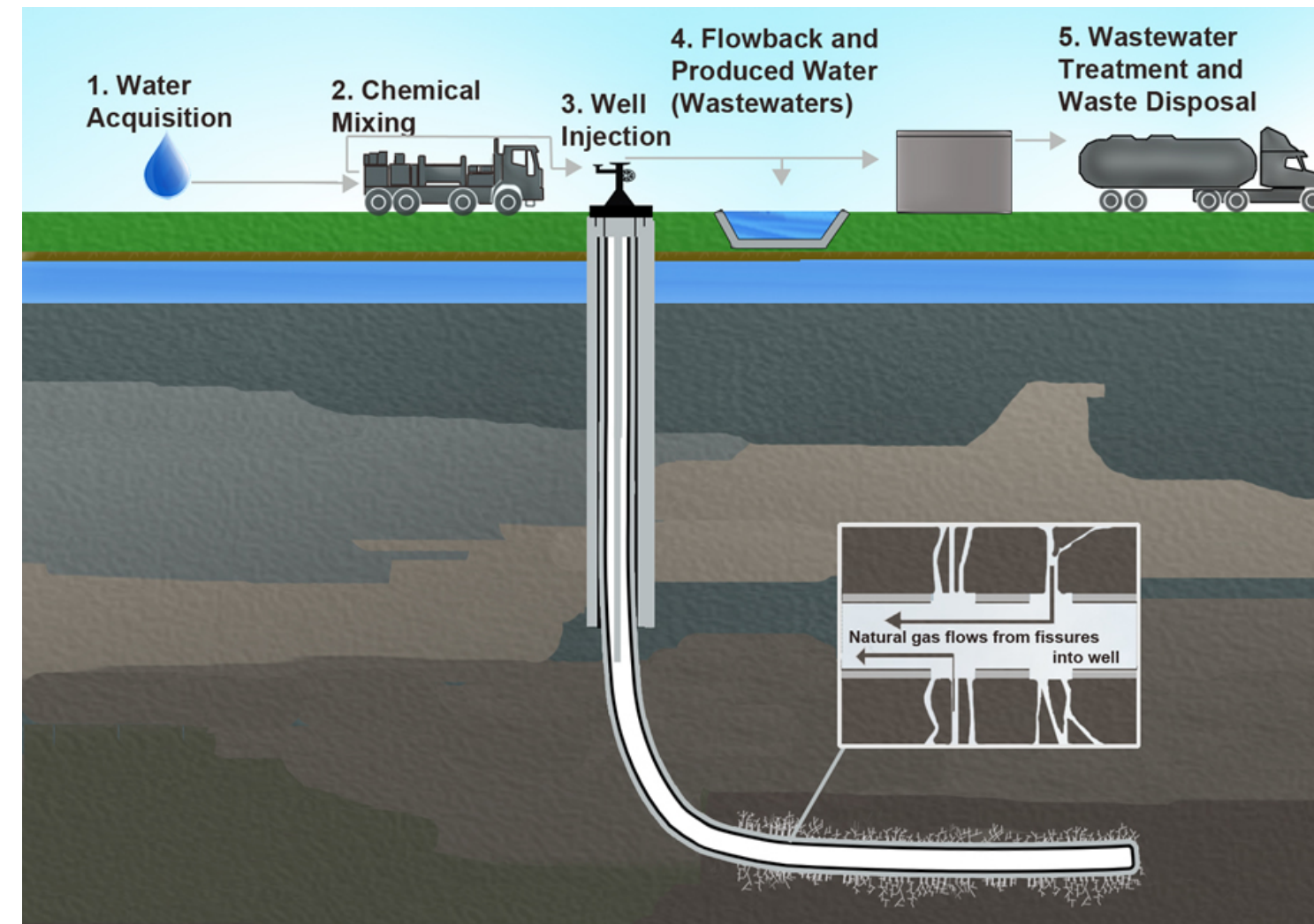
Owen, Chapters 1 & 2

Assignment 1 Due Sep 5

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025



Monte Carlo Helps With Uncertainty [WSJ 2016]





Why is there uncertainty?



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- Neural networks — many parameters need to be tuned, but one cannot search in **all possible directions**
- Queues — **arrival times and service times** of customers



How is this expressed quantitatively?

Y = random variable denoting quantity of interest = {
option payoff
fluid pressure
pixel intensity
statistical model parameter
neural network parameter
service time

= $f(\mathbf{X})$, where

$\mathbf{X}$ = multivariate random variable with a simpler distribution

We want to estimate the mean, variance, quantile, or probability distribution of Y
using sample versions



Are we there yet?

You are visiting your friend and it will require

- A 5 minute walk to the 'L' station
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 - There is a 20% chance that the car is waiting for you
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How long should you plan for the trip to take?

Let's look at this Jupyter Notebook [AreWeThereYet](#) on the [class website](#)



Let me know your degree program?

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Why should you attend synchronously?

- You will be better keep **pace**?
- You will get **real time** answers to questions?
- You can **influence** the pace and direction of the course?
- You can help your peers learn and benefit from them—partake in a **learning community**
- To help me know you, in case you want a **reference**, or to add me to your **LinkedIn** network



Review of probability

Let Y be a **random variable** with a sample space $\mathcal{Y} \subseteq \mathbb{R}$

$\mathbb{P}(Y \in \Omega)$ means the probability of the **event** Ω

F is the **cumulative distribution function** $F(y) := \mathbb{P}(Y \leq y)$, $y \in \mathcal{Y}$

Q is the **quantile function** $Q(p) := \inf\{y \in \mathcal{Y} : F(y) \geq p\}$, $0 < p < 1$

$\mathbb{E}[g(Y)] := \int_{\mathcal{Y}} g(y) \, dF(y)$ is the **expectation** of $g(Y)$

discrete

continuous

ϱ is the **probability mass function**

$$\varrho(y) := \mathbb{P}(Y = y)$$

$$\mathbb{E}[g(Y)] = \sum_{y \in \mathcal{Y}} g(y) \varrho(y)$$

ϱ is the **probability density function**

$$\varrho(y) := F'(y)$$

$$\mathbb{E}[g(Y)] = \int_{y \in \mathcal{Y}} g(y) \varrho(y) \, dy$$



Moments

$\mu := \mathbb{E}(Y)$ is the **mean** of Y

$\sigma^2 = \text{var}(Y) := \mathbb{E}[(Y - \mu)^2]$ is the **variance** of Y

σ is the **standard deviation**

$$\text{cov}(\mathbf{Y}) := \left(\mathbb{E}[(Y_j - \mu_j)(Y_k - \mu_k)] \right)_{j,k=1}^d$$

is the **covariance matrix** of the random **vector** $\mathbf{Y}$



Binomial Distribution

This is a coin-flipping distribution with the probability p of heads

$$Y \sim \text{Binomial}(n, p) \quad \text{scipy.stats.binom}(n, p)$$

$$\varrho(y) := \mathbb{P}(Y = y) = \binom{n}{y} p^y (1 - p)^{n-y}, \quad y = 0, 1, \dots, n$$

$$F(y) = \mathbb{P}(Y \leq y) = \sum_{j=0}^{\lfloor y \rfloor} \binom{n}{j} p^j (1 - p)^{n-j}$$

$$Q(p) = \inf\{y \in \{0, 1, \dots, n\} : F(y) \geq p\}$$

$$\mu = \mathbb{E}(Y) = np \quad \sigma^2 = \text{var}(Y) = np(1 - p)$$



Multivariate normal/Gaussian distribution

Used in finance and uncertainty quantification

$Y \sim \mathcal{N}(\mu, \Sigma)$ is a d -dimensional random vector

`scipy.stats.multivariate_normal(mean, cov)`

μ = mean

Σ = covariance matrix

$$\varrho(\mathbf{y}) := \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mu)^\top \Sigma^{-1}(\mathbf{y} - \mu)\right)$$

$$F(\mathbf{y}) = \int_{(-\infty, \mathbf{y})} \varrho(\mathbf{z}) \, d\mathbf{z}$$



Sampling

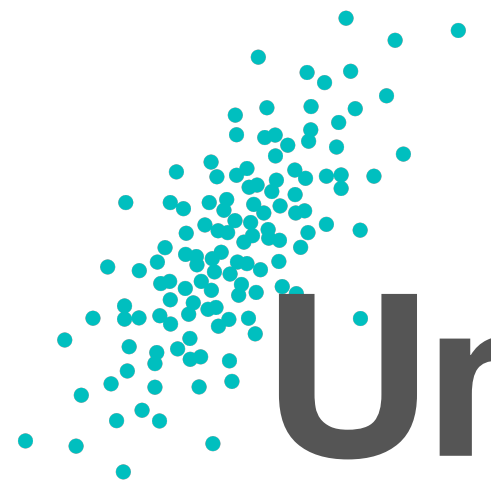
Let Y be a random variable, often $Y = f(\mathbf{X})$

$Y_1, Y_2, \dots$ be a **sample**

not necessarily random or independent and identically distributed (IID)

$\hat{\mu}_n := \frac{1}{n} \sum_{i=1}^n Y_i$ is the **sample mean**

$\hat{\sigma}_n^2 := \frac{1}{n-1} \sum_{i=1}^n (Y_i - \hat{\mu}_n)^2$ is the **sample variance**



Under **random** sampling

Let Y be a random variable, often $Y = f(\mathbf{X})$

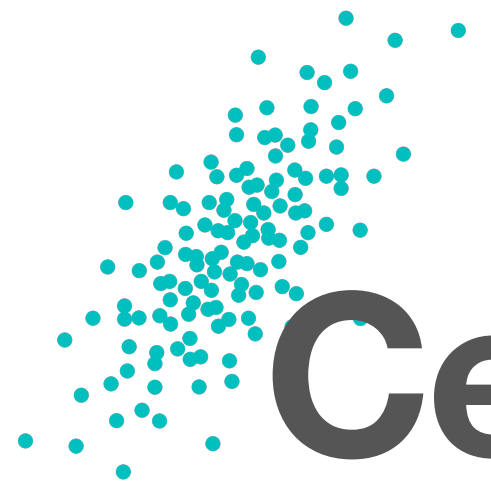
$$Y_1, Y_2, \dots \sim Y$$

$$\mathbb{E}(\hat{\mu}_n) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(Y_i) = \mu \text{ so } \hat{\mu}_n \text{ is unbiased}$$

If in addition, $Y_1, Y_2, \dots$ are **uncorrelated**, then

$$\mathbb{E}(\hat{\sigma}_n^2) = \sigma^2 \text{ so } \hat{\sigma}_n^2 \text{ is unbiased}$$

$$\text{mse}(\hat{\mu}_n) := \mathbb{E}[(\mu - \hat{\mu}_n)^2] = \text{var}(\hat{\mu}_n) = \frac{\sigma^2}{n}, \quad \text{rmse}(\hat{\mu}_n) = \frac{\sigma}{\sqrt{n}}$$



Central Limit Theorem

Let Y be a random variable, often $Y = f(\mathbf{X})$

$$Y_1, Y_2, \dots \stackrel{\text{iid}}{\sim} Y$$

The distribution of

$$\frac{\hat{\mu}_n - \mu}{\sigma / \sqrt{n}} \text{ approaches } \mathcal{N}(0, 1) \text{ as } n \rightarrow \infty$$

which means that

$$\hat{\mu}_n \dot{\sim} \mathcal{N}(\mu, \sigma^2/n) \text{ for large } n$$



Monte Carlo estimates of properties of Y

Let Y be a random variable and let $Y_1, Y_2, \dots$ be an IID random sample

Population Quantity	Sample Approximation
mean μ	$\hat{\mu}_n \pm 2.58\hat{\sigma}_{n_0}^2$, <code>np.mean(data)</code>
variance σ^2	$\hat{\sigma}_n^2$, <code>np.var(data, ddof=1)</code>
cumulative distribution function F	empirical distribution function $\hat{F}_n$ <code>statsmodels.distributions.empirical_distribution.ECDF(data)</code>
quantile function Q	empirical quantile function $\hat{Q}_n$ <code>np.quantile(data, probabilities)</code>
probability density or mass function ϱ	histogram, <code>np.histogram(data, bins = "auto")</code> kernel density estimator, <code>scipy.stats.gaussian_kde(data)</code>

For μ we have an **interval** estimator, but for the others typically **point** estimators



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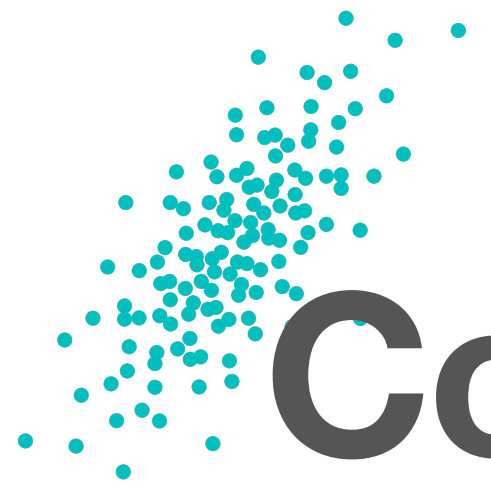
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Conditional probability

Let (Y, Z) be a vector of random variables

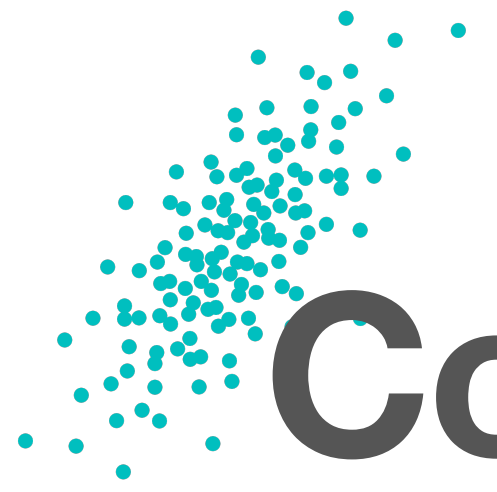
$$\mathbb{P}(Y = y \mid Z = z) = \frac{\mathbb{P}(Y = y \& Z = z)}{\mathbb{P}(Z = z)}$$

$$\mathbb{P}(Y = y \mid Z = z) = \frac{\mathbb{P}(Z = z \mid Y = y)\mathbb{P}(Y = y)}{\mathbb{P}(Z = z)} \quad \text{Bayes' Theorem}$$

$$\varrho_{Y|Z}(y \mid z) = \frac{\varrho_{Y,Z}(y, z)}{\varrho_Z(z)}$$

$$\mathbb{E}(Y) = \mathbb{E}_Z[\mathbb{E}_Y(Y \mid Z)]$$

$$\text{var}(Y) = \mathbb{E}_Z[\text{var}_Y(Y \mid Z)] + \text{var}_Z[\mathbb{E}_Y(Y \mid Z)]$$



Conditional Monte Carlo

Let $(Y = f(\mathbf{X}_{1:d}), \mathbf{X}_{1:d})$ be a vector of random variables

Let $\mathbf{Z} = \mathbf{X}_{2:d}$

If $g(\mathbf{X}_{2:d}) = \mathbb{E}_Y(Y \mid \mathbf{X}_{2:d})$ has an analytic form, then

$$\mu = \mathbb{E}(Y) = \mathbb{E}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})] \text{ and } \text{var}(Y) \geq \text{var}_Z[g(\mathbf{X}_{2:d})]$$

If $h(y, \mathbf{X}_{2:d}) = \varrho_{Y \mid \mathbf{X}_{2:d}}(y \mid \mathbf{x}_{2:d})$ has an analytic form, then

$$\varrho_Y(y) = \mathbb{E}_{\mathbf{X}_{2:d}}[h(y, \mathbf{X}_{2:d})] \quad \text{probability density expressed as expectation}$$



Big Ideas

- **Sample quantities**, often using IID samples, used to estimate population properties of a random variable with a complex distribution, including
 - Means, variances, covariances
 - Probability density/mass functions, cumulative distribution functions, and quantile functions
- For means we have **interval** estimators, which indicate the uncertainty in our estimates, using the CLT
- For the other properties we have **point** estimators, i.e., only one approximate value for each thing to be estimated